

Scenario 1: Delivery Time Analysis for an E-commerce Company

An e-commerce company tracks delivery times (in minutes) for 15 orders:
[25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95]

The company wants to analyze the delivery performance using percentiles and detect if there are any unusual delivery times.

Question 1:

Calculate Q1 and Q3.

Q1 - 25th % of Data

Formula : $0.25(n+1)$. $n=15$. $0.25 \times 16 = 4$

4th value = 40

Q3 - 75th % of Data

Formula : $0.75(n+1)$. $n=15$. $0.75 \times 16 = 12$

12th value = 80

Question 2:

Find the Interquartile Range (IQR).

$IQR = Q3 - Q1 = 80 - 40 = 40$

Question 3:

Detect Outliers using the IQR method.

Lower Bound = $Q1 - IQR \times 1.5 = 40 - 60 = -20$

Upper Bound = $Q3 + IQR \times 1.5 = 80 + 60 = 140$

Numbers lesser than Lower Bound and higher than Upper Bound are the Outliers.

This dataset has not Outliers.

Scenario 2: Student Score Analysis

A teacher is analyzing the mathematics scores of students in her class. The scores are:

[45, 50, 55, 60, 60, 62, 63, 65, 90, 95]

Question 1:

Calculate the mean, median, and mode of the scores.

Answer 1:

Mean (average):

Mean = sum of all values / $n = 10$

Sum = $45 + 50 + 55 + 60 + 60 + 62 + 63 + 65 + 90 + 95 = 645$

Mean = $645 / 10 = 64.5$

Median = average of 5th and 6th values. 5th = 60, 6th = 62

$(60 + 62) / 2 = 61$

Mode = (most frequent value):

60 appears twice, all others once.

So Mode = **60**

Question 2:

Explain why the median might be a better representation than the mean in this case.

Answer 2:

The median (61) is a better representation than the mean (64.5) because most scores are clustered around 50–65, and the mean is pulled up by the high outliers (90, 95). Median shows the typical middle value without being affected by extreme scores.

Scenario 3: Grocery Store Customer Analysis`

A grocery store manager tracks how many customers visit the store daily for a month:

[5, 10, 8, 15, 20, 5, 12, 14, 10, 18]

Question 1:

Create a frequency distribution table for this data.

No of Customers	Frequency
5	2
8	1
10	2
12	1
14	1
15	1
18	1
20	1

Scenario 4: Real Estate Model Analysis

A real estate model has three variables:

- House Size
- Number of Rooms
- Number of Bathrooms

Question 1:

How can you detect multicollinearity in this model?

Logic: Calculate Variance Inflation Factor (VIF)

Rule of thumb: $VIF > 5$ or $10 \rightarrow$ high multicollinearity.

Scenario 5: Medicine Effectiveness Study

A company made a new medicine to lower blood pressure. They gave it to one group and gave a fake pill (placebo) to another group.

Question 1:

How can the company check if the new medicine works?

Answer:

State Hypotheses:

- Null Hypotheses : There is no significant difference between the results from 2 groups - New Medicine does NOT work
- Alternative Hypotheses : There is a significant difference between the results from 2 groups - New Medicine works.

Do T-Test:

- Calculate p-value and check if it is less than 0.05

Conclusion:

- If p-value is less than 0.05, the New Medicine works.

Scenario 6: Identifying Outliers in Sales Data

A company wants to find any unusual spikes in sales.

Question 1:

How can the company detect outliers in their sales data?

Answer:

1. Calculate Q1 and Q3 for the data.
2. Calculate the IQR = $Q3 - Q1$
3. Calculate $1.5 \times \text{IQR}$
4. Find values in Lower Bound = $Q1 - 1.5 \times \text{IQR}$ and Upper Bound = $Q3 + 1.5 \times \text{IQR}$
5. Outliers are value less than Lower Bound and higher than Upper Bound.

Scenario 7: Understanding Customer Satisfaction

A restaurant conducted a survey to rate customer satisfaction on a scale of 1 to 5:
[5, 4, 4, 5, 3, 4, 5, 2, 4, 3]

Question 1:

How can the restaurant summarize the overall satisfaction?

Answer :

- Mean = $(5+4+4+5+3+4+5+2+4+3) / 10 = 3.9$
- Median = middle of sorted data [2, 3, 3, 4, 4, 4, 4, 5, 5, 5]
 $(5^{\text{th}}+6^{\text{th}})/2 = (4+4)/2 = 4$
- Mode = most frequent = 4

Most customers rated the restaurant highly, with an average score of 3.9, a median of 4, and mode of 4, showing overall satisfaction is "good."